



SCHOOL OF SCIENCE & TECHNOLOGY: DEPARTMENT OF MATHEMATICS

SEMESTER: Fall 2024

COURSE: DSA 1080 A- Programming for Data Science

TASK: GROUP PROJECT 1

INSTRUCTOR: DR. MAURINE WAFULA **Maximum: 20 Marks**

DEADLINE: 12/11/2024

ATTEMPT ALL THE QUESTIONS AND SUBMIT ON TIME

Honey Production Case Study

Background

In 2006, global concern was raised over the rapid decline in the honeybee population, an integral component of American honey agriculture. Large numbers of hives were lost to Colony Collapse Disorder, a phenomenon of disappearing worker bees that causes the remaining hive colonies to collapse. Speculation on the cause of this disorder points to hive diseases and pesticides harming the pollinators, though no overall consensus has been reached. The U.S. used to locally produce over half the honey it consumes per year. Nowadays, honey mostly comes from overseas, with 350 of the 400 million pounds of honey consumed every year originating from imports. This dataset provides insight into honey production supply and demand in America from 1998 to 2016.

Objective:

To visualize how honey production has changed over the years (1998–2016) in the United States.

Key questions to be answered:

- How has honey production yield changed from 1998 to 2016?
- Over time, what have been the major production trends across the states?
- Are there any pattern that can be observed between total honey production and the value of production every year? How has the value of production, which in some sense could be tied to demand, changed every year?

Dataset:

- **State:** Various states in the U.S.
- **year:** Year of production
- **stocks:** Refers to stocks held by producers. Unit is pounds
- **numcol:** Number of honey-producing colonies. Honey producing colonies are the maximum number of colonies from which honey was taken during the year. It is possible to take honey from colonies that did not survive the entire year
- **yieldpercol:** honey yield per colony. The unit is in pounds
- **totalprod:** Total production (numcol x yieldpercol). Unit is pounds
- **priceperlb:** Refers to average price per pound based on expanded sales. The unit is dollars.
- **prodvalue:** Value of production (totalprod x priceperlb). The unit is dollars.

1.Import the necessary packages

2. Read the datasets(CODE) and give insight

3. Displaying the first few rows of the dataset

5. shape of the user dataset (CODE) and give insight

6. Checking the datatype of the columns (CODE) and give insight

7. Get the statistical summary for the dataset (CODE) and give insight

8. look at the spread of number of honey-producing colonies using

i. Histogram

ii. Box plot

In both cases. CODE and give insight

9. look at the spread of yield per colony using

i. Histogram

ii. Box plot

In both cases. CODE and give insight

10. look at the distribution of total production of honey using

i. Histogram

ii. Box plot

In both cases. CODE and give insight

11. look at the distribution of cost of honey per pound

i. Histogram

ii. Box plot

In both cases. CODE and give insight

12. look at the distribution of value of production.

i. Histogram

ii. Box plot

In both cases. CODE and give insight

13. look at the distribution of stocks held by producers.

i. Histogram

ii. Box plot

In both cases. CODE and give insight

14. visualise how these numerical variables are associated with each other. CODE and give insight

15. create a 2-D Matrix with correlation plots

ii. create heatmap with cmap="Spectral" and values inside. Give insight

16. visualise trends in columns state and year give insight

17. extract which states have maximum and minimum honey production

18. create a catplot for top10_totalprod, x= 'state', y='totalprod', of kind=bar

19. create top 10 states producing minimum honey bottom10_totalprod

20. create catplot(data=bottom10_totalprod, x= 'state', y='totalprod', of kind=bar

21. check the which state is producing costliest and cheapest honey on an average basis using the code below

```
top10_price=
honeyprod.groupby('state').sum()[['priceperlb']].sort_values('priceperlb',
ascending=False).reset_index().head(10) #top 10 states producing maximum
honey
top10_price
```

22. create catplot(data=top10_price, x= 'state', y='priceperlb', kind='bar' with plt.xticks at 90 in figsize=(20,6)

23. check top 10 states producing minimum honey bottom10_price using the code below

```
bottom10_price=
honeyprod.groupby('state').sum()[['priceperlb']].sort_values('priceperlb',
ascending=False).reset_index().tail(10)
```

24. plt.figure()

25. Plot catplot of bottom10_price, x= 'state', y='priceperlb, with kind=bar and plt.xticks rotation=90, with figsize=(20,6)

Variation in the number of colonies over the years

26. Plot pointplot (x='year', y='numcol', data=honeyprod, ci=None, estimator=sum) plt.xticks(rotation=90), with figsize=(15, 7)

Variation of yield per colony over the years

26. Plot pointplot(x='year', y='yieldpercol', data=honeyprod, estimator=sum, ci=None) plt.xticks(rotation=90), with figsize=(15, 7)

production trend at state level

27. use the code below

```
# Add hue parameter to the pointplot to plot for each state
plt.figure(figsize=(15, 7)) # To resize the plot
sns.pointplot(x='year', y='totalprod', data=honeyprod, estimator=sum,
ci=None, hue = 'state') plt.legend(bbox_to_anchor=(1, 1))
plt.xticks(rotation=90);
```

28. Plot catplot(x='year', y='totalprod', data=honeyprod, estimator=sum, col='state' with kind=point, col_wrap = 5);

yearly trend in number of colonies and yield per colony in the 5 prominent states

29. Use the code below

```
cplot1=sns.catplot(x='year', y='numcol',
data=honeyprod[honeyprod["state"].isin(["North Dakota","California","South
Dakota","Florida","Montana"])], estimator=sum, col='state', kind="point",
col_wrap = 5) cplot1.set_xticklabels(rotation=90);
```

30. Use the code below

```
cplot2=sns.catplot(x='year', y='yieldpercol',
data=honeyprod[honeyprod["state"].isin(["North
Dakota","California","South Dakota","Florida","Montana"])],
estimator=sum, col='state', kind="point",
col_wrap = 5)
cplot2.set_xticklabels(rotation=90);
```

effect, the declining production trend has on the value of production

31. Use the code below

```
sns.pointplot(x="year", y="prodvalue", data=honeyprod, ci=None)
plt.xticks(rotation=90);
```

check which of the states have been capitalising on this trend. We can compare the total production with the stocks held by the producers.

32. Use the code below

```
plt.figure(figsize = (20,15)) # To resize the plot

# Plot total production per state
sns.stripplot(x="state", y="totalprod",
data=honeyprod.sort_values("totalprod", ascending=False),
              label="Total Production", color="b", jitter=True)
plt.xticks(rotation=90);
```

32. Use the code below

```
plt.figure(figsize = (20,15)) # To resize the plot
# Plot stocks per state
sns.stripplot(x="state", y="stocks", data=honeyprod.sort_values("totalprod",
ascending=False),
              label="Stocks", color="r", jitter=True)

plt.xticks(rotation=90);
```

average price per pound of honey across states

33. Use the code below

```
plt.figure(figsize=(25, 7)) # To resize the plot
sns.swarmplot(data = honeyprod, x = "state", y = "priceperlb",
              )
plt.xticks(rotation=90);
```

34. Give conclusion

